

Assignment

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1 Question 1

The following Technical Debts are seen in the given scenarios in order:

- (a) Entanglement
- (b) Undeclared Consumers
- (c) Configuration & Glue-Code Debt

To address Configuration and Glue-Code Debt, pin the code and data via Git and DVC, then instrument the run using tools like MLflow.

Having a smaller batch size should not inherently lead to better results, which is evident as the accuracy only differs in the first decimal place.

3 Question 3

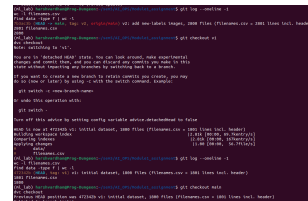


Figure 2: Number of rows before and after rollback

2 Question 2

	lr	batch	epochs	train loss	val acc	best val	test acc
	0.001	32	15	0.0103	0.9631	0.9631	0.9627
	0.001	64	15	0.0131	0.9563	0.9681	0.9575
	0.001	128	15	0.0113	0.9663	0.9675	0.9610
	0.01	32	15	0.1002	0.9419	0.9494	0.9445
	0.01	64	15	0.0691	0.9450	0.9550	0.9385
	0.01	128	15	0.0412	0.9587	0.9625	0.9520

Best config: lr=0.001 batch_size=32 -> test accuracy=0.9627

Figure 1: Results from the ML training run

It is clear that the best set of parameters happens to be a learning rate of 0.001 and a batch size of 32. Clearly, having a smaller learning rate was the more significant factor compared to batch size, but both led to a significant shift in test accuracy.

A lower learning rate would have helped the model reach narrow cusps better than a large learning rate, which can lead to overshoots and hence revolve around the same point. Both experiments with a batch size of 64 have a high drop in best validation accuracy and test accuracy, suggesting that they overfitted to the validation set the most.

4 Question 4

Please access repository: Repo